

Course Outline

Data & Business Process Modeling

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Slides And Videos - Download, Watch, Interact



Watch this lecture and download the slides from

<http://jarrar-courses.blogspot.com/2015/01/dataandbusinessprocessmodelling.html>



Course Webpage

<http://jarrar-courses.blogspot.com/2015/01/data-and-business-process-modelling.html>

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Data & Business Process Modelling Course



Course Outline

This course aims to widen students' understanding and enhance their modeling skills on how conceptualize and engineer data, business rules, and business processes. The first module focuses on (i) Conceptual data modeling: conceptual analyses using Object Role Modelling ORM, business rules, integrity and derivation rules, contradictions and implications between rules, conceptual schema optimization and engineering, business rules verbalization. The second module focuses on (ii) Business Process Management: Concepts and principles, modeling notations and languages, process identification, optimization and (re-)engineering, and Process implementation.

Part I : Conceptual Data Modeling using ORM:



Concepts Principles of Conceptual Data Modeling

By: Mustafa Jarrar



Conceptual Schema Design Steps

By: Mustafa Jarrar



Uniqueness

By: Mustafa Jarrar



Mandatory Roles

By: Mustafa Jarrar



Other Constraints

By: Mustafa Jarrar



Final Check Schema Engineering

By: Mustafa Jarrar

Part II : Business Process Modelling

Course Readings

- Mustafa Jarrar: **Lecture Notes on Data and Business Process Modeling**, Birzeit University. <http://jarrar-courses.blogspot.com/2015/01/dataandbusinessprocessmodelling.html>
- Terry Halpin, Tony Morgan: **Information Modeling and Relational Databases**, Second Edition. Second Edition. The Morgan Kaufmann Series in Data Management Systems. ISBN: 0123735688
- Mathias Weske: **Business Process Management: Concepts, Languages, Architectures**. Second Edition, Springer. ISBN: 978-3642286155
- Walter T. Geer Jr.: **What Lucy Taught Us: A Management Fable about Improving Your Business One Process at a Time**. BookSurge Publishing. ISBN: 978-1439243213

Keywords: Data Engineering, Data Modeling, Conceptual Data Modeling, ORM, Object Role Modeling, Information Modeling, Schema engineering, Business Rules, integrity constraints, Derivation rules, Logic, verbalization of business rules, rules contradictions, rules implications, Business Process Engineering, Business Process Management, Business Process Modelling, Process Mapping, Process identification, process re-engineering, process re-engineering, Process Optimization, Process execution.

مندسة البيئات، مخطط بيئات المفاهيمية، النمذجة المفاهيمية للبيئات، هندسة العمليات، مندر

العمليات، إعادة هندسة العمليات الإدارية mjarrar © 2015

Course Outline



The course is divided into two parts:

Part I: Conceptual Data Modeling, using ORM

Part II: Business Process Management

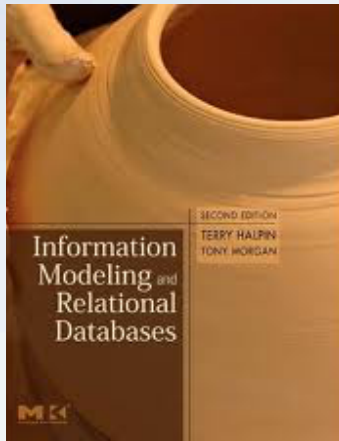
There is no perfect design, there is a better design!

Help you to learn science and art!

➔ Better salary, better communication with people

Part I: Conceptual Data Modeling, using ORM

Topics



Terry Halpin, Tony Morgan: **Information Modeling and Relational Databases**, Second Edition. Second Edition. The Morgan Kaufmann Series in Data Management Systems. ISBN: 0123735688

	Information Modeling and Conceptual Analyses
	Project-1
	Uniqueness and Mandatory Rules
	Project-2
	Subtype Relations, and Other business rules
	Schema Equivalence, Optimization, and Engineering
	Project-3
	Medterm Exam

Learning Objectives (Part I)

Knowledge and Understanding

- *Notations and concepts* used in conceptual modeling
- *Object Role Modeling (ORM) Methodology.*
- *Data integrity and business rules.*

Intellectual Skills:

- Analyze application requirements at the conceptual level, and formalize it using ORM.
- Analyze entity identities
- Map conceptual models into DB databases & other models.
- Optimize/transform/(re-)engineer conceptual models.
- Detect/resolve contradictions and implications

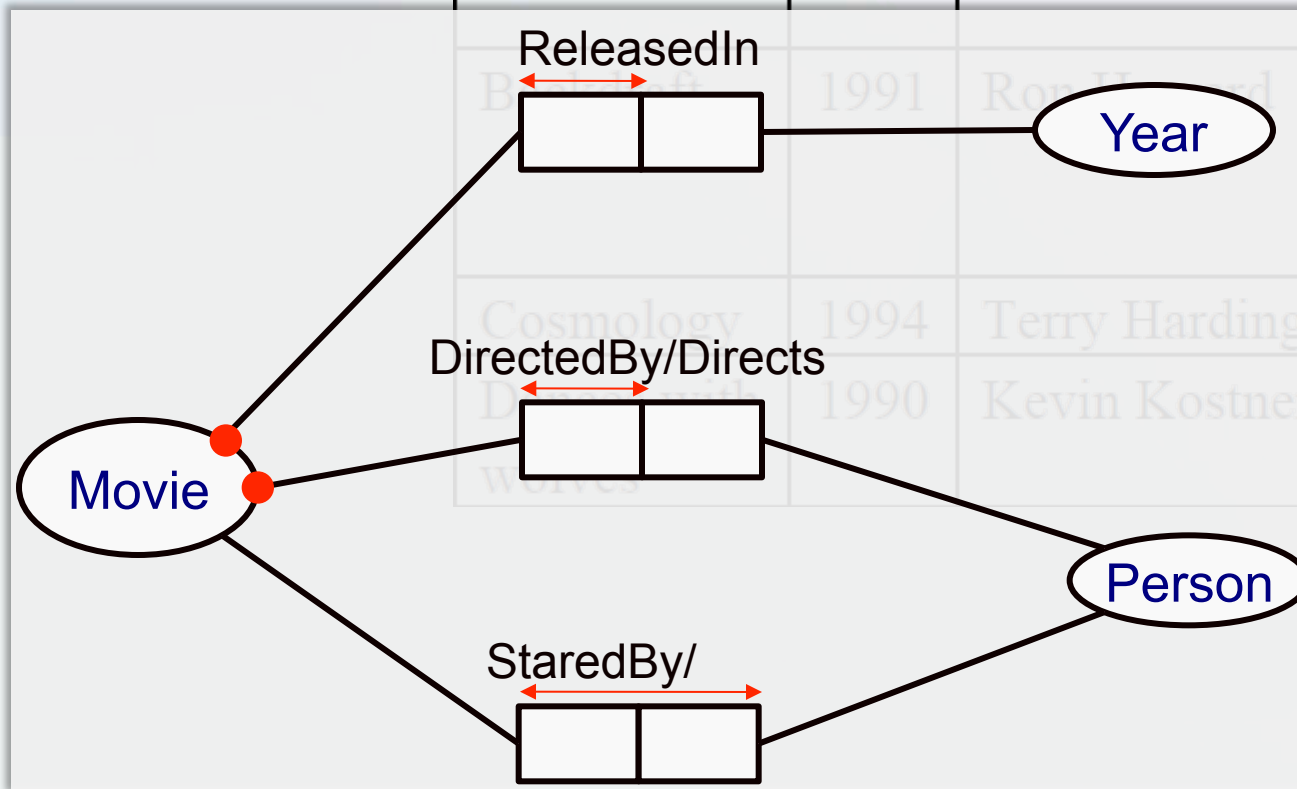
Professional and Practical Skills:

- Use ORM modeling tools, such as NORMA.
- Generate databases normalized at the 5th normal form

Object-Role Modeling (ORM)

Representing information graphically

<i>Movie</i>	<i>Year</i>	<i>Director</i>	<i>Stars</i>
Awakenings	1991	Penny Marshall	Robert De Niro Robin Williams
			William Baldwin Robert De Niro Kurt Russell
			Kevin Kostner Mary McDonnell



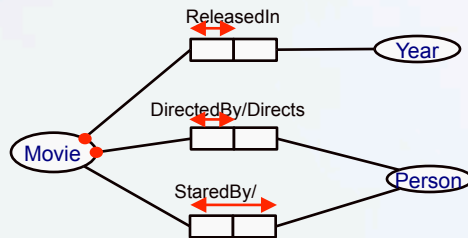
Object-Role Modeling (ORM)

ORM is conceptual modeling language.

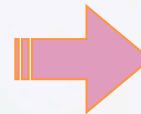
ORM has an expressive graphical notation.

ORM is designed for modeling DB schemes at the conceptual level.

You build an ORM schema and then click a button to automatically generate a database.



Conceptual Level



MovieName	Release Year	Director
Awakenings	1991	Penny Marshall
Backdraft	1991	Ron Howard
Cosmology	1994	Terry Harding
Dances with wolves	1990	Kevin Kostner

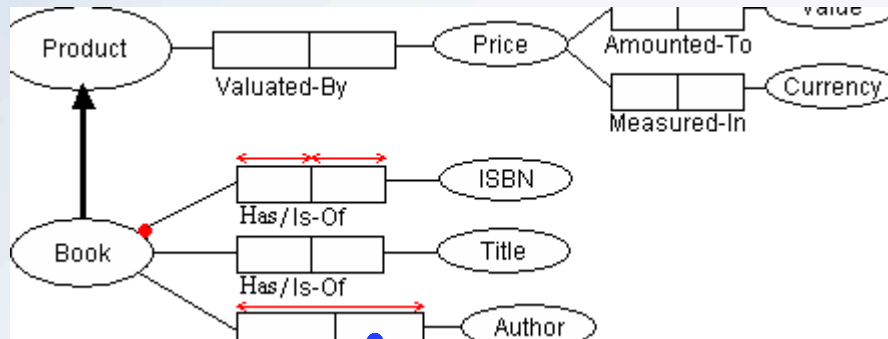
MovieName	Star
Awakenings	Robert De Niro
Awakenings	Robin Williams
Backdraft	William Baldwin
Backdraft	Robert De Niro
Backdraft	Kurt Russell
Cosmology	
Dances with wolves	Kevin Kostner
Dances with wolves	Mary McDonnell

Logical Level

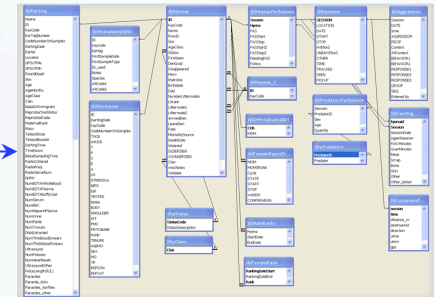
➔ Our goal in this course is to use ORM as general Conceptual Modeling language, rather than only as database modeling language.

➔ ORM can be used for modeling business rules, ontology, XML schemes, and others.

ORM Usage Scenarios



Originally



Database

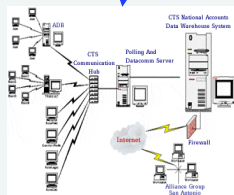
Later



Ontology



XML Schema



Warehouse



Business Rules



Web (x)Forms

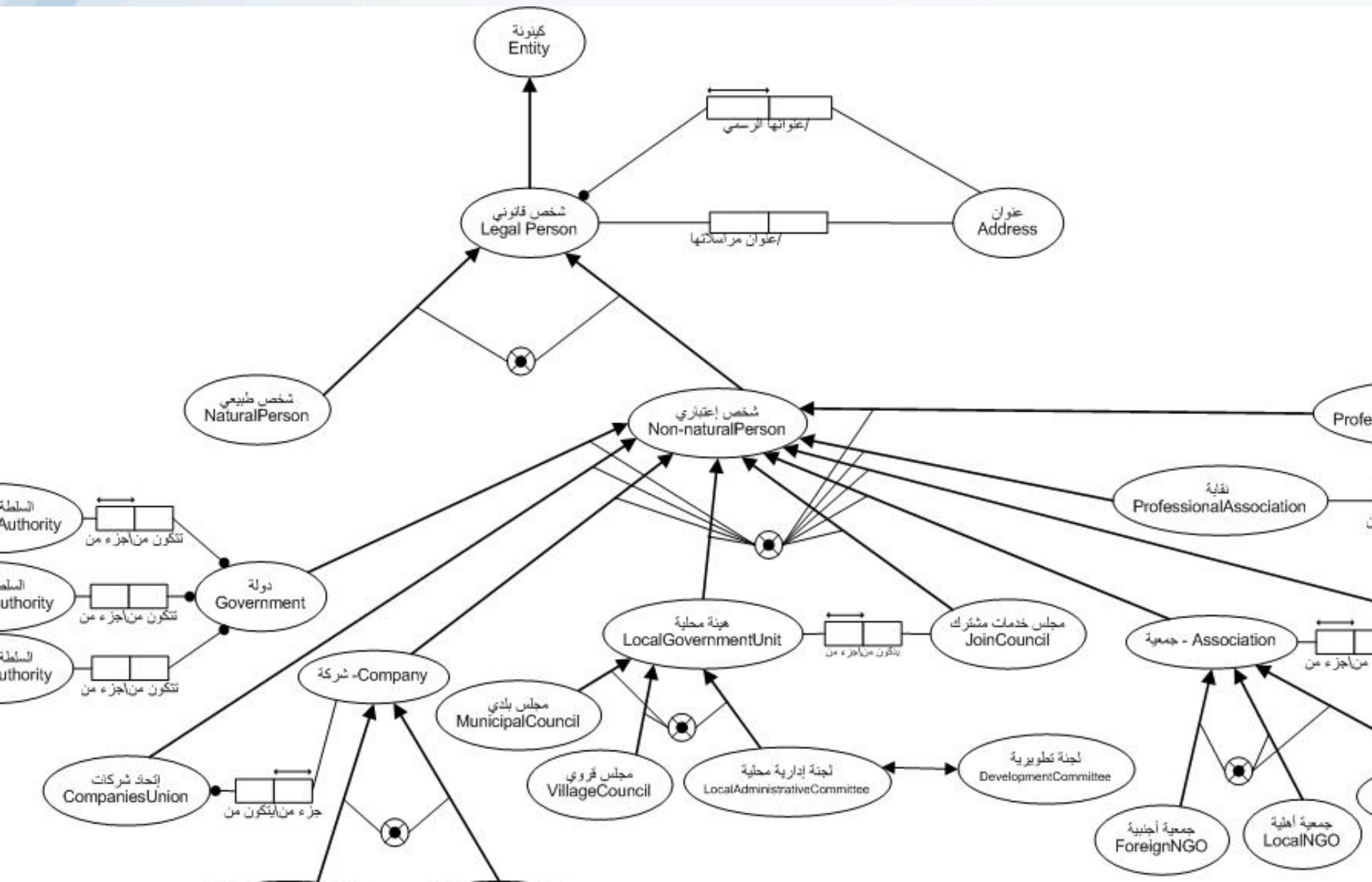


Requirements Engineering



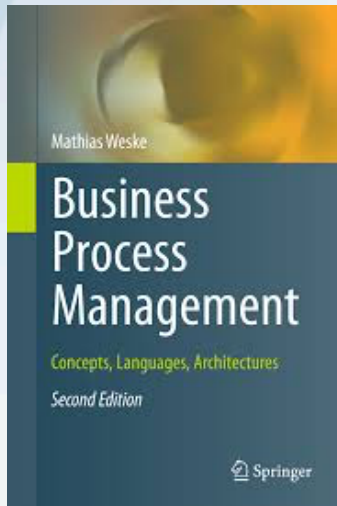
Record my recipes !

eGov Ontology (using ORM)



Part I: Business Process Modeling

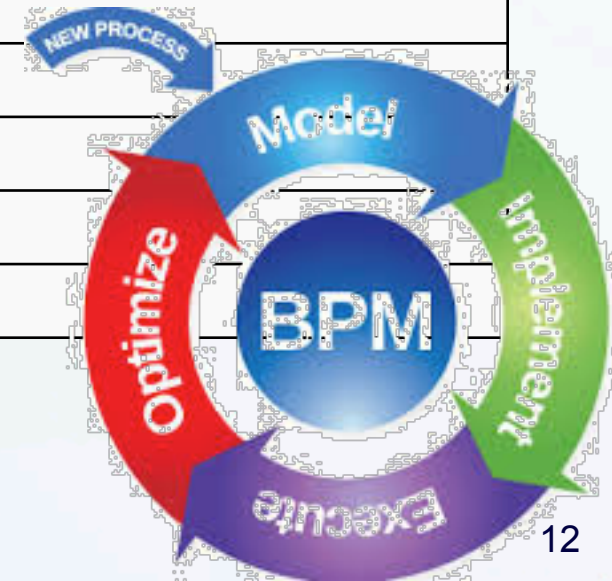
Topics



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	Business Process Analyses
	Business Process Languages and notations
	Project-1
	Business Process improvements
	Project-2
	Business Process implementation
	Project-3



Learning Objectives (Part II)

Knowledge and Understanding

- *business process modeling and mapping.*
- *business process optimization and (re-)engineering.*

Intellectual Skills:

- *Model and map business processes (BPMN 2.0)*
- *Optimize and re-engineer business processes.*

Professional and Practical Skills:

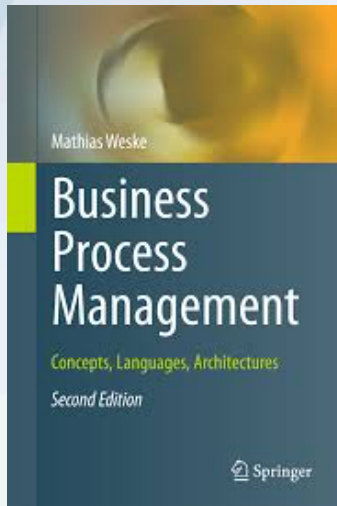
- *Use process modeling tools, such as Activiti, Visio and Piazzi.*
- *Implement and execute process models using JAVA frameworks, and ingrate in service oriented Architecture.*

Other and Soft Skills:

- *Communicate with domain experts and business owners.*
- *Deliver technical presentations.*
- *Work in teams.*

Part I: Business Process Modeling

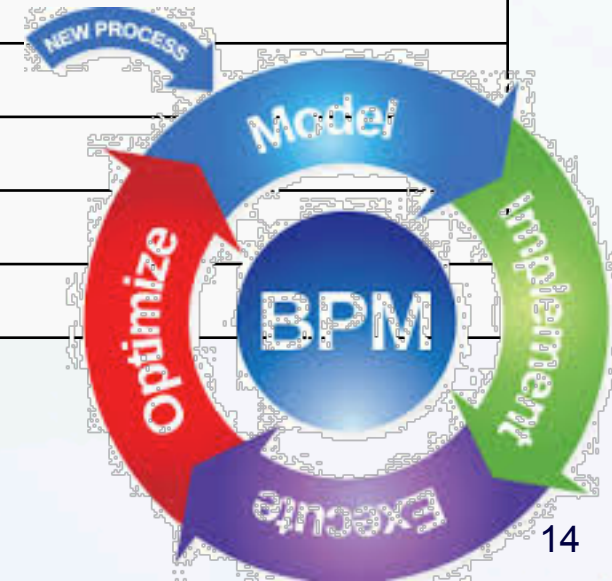
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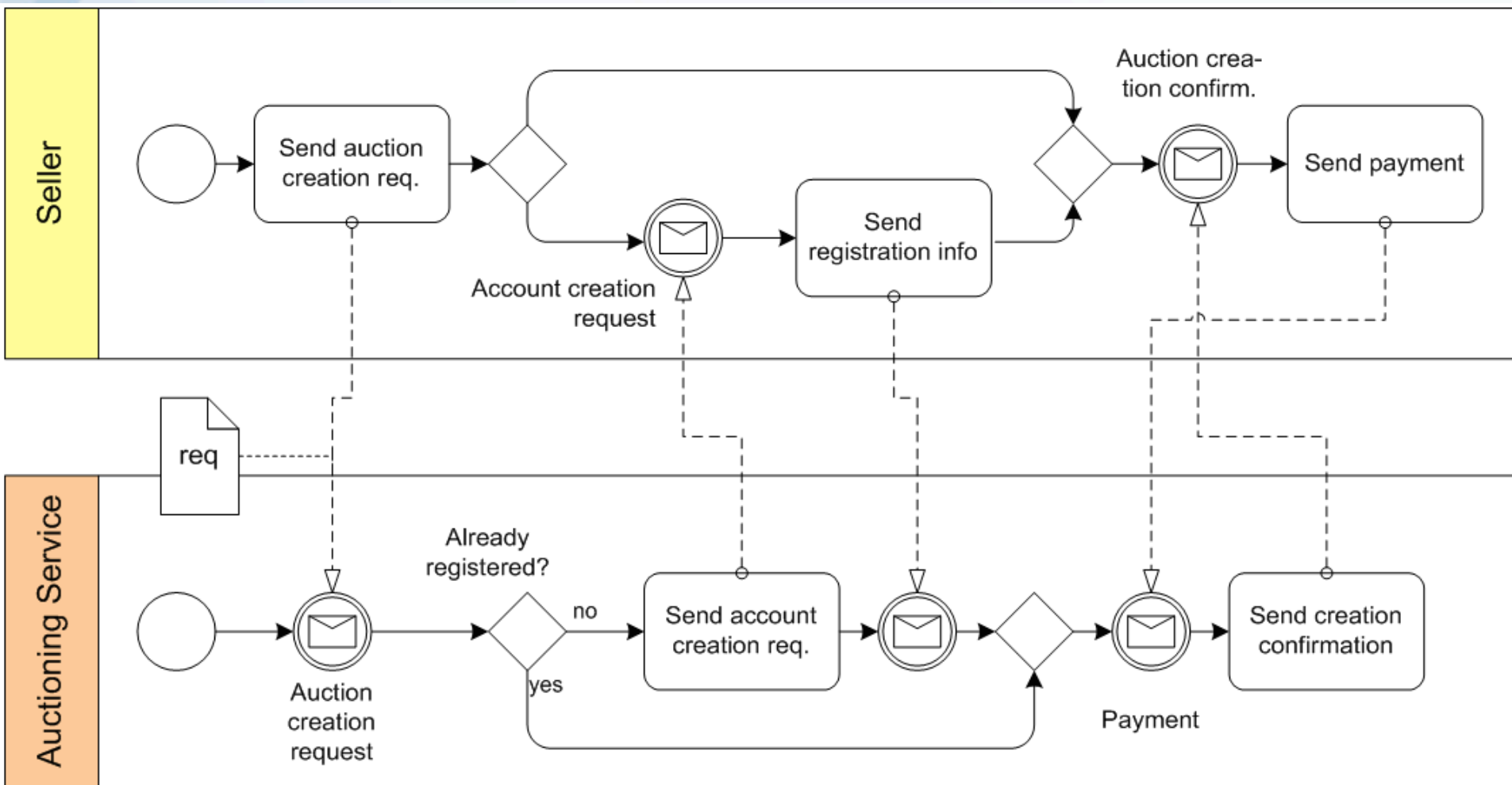
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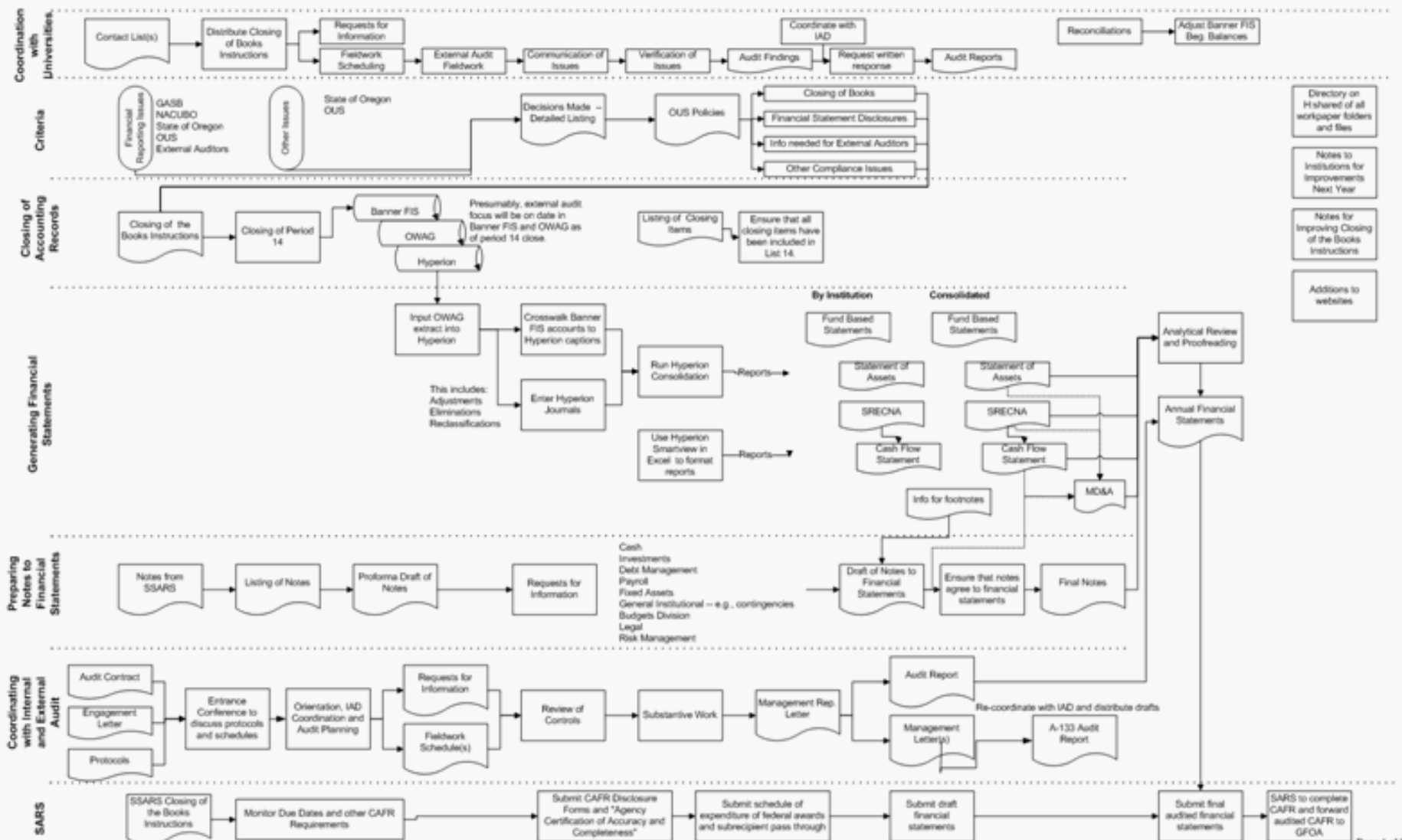


Example



Example

General Flowchart of Financial Reporting Process.vsd



Evaluation

Midterm exam	20%
Assignments	10%
Projects	35%
Interaction & contribution	5%
Final Exam	30%

Students Responsibility

Bing your laptop: As this is a practical oriented course, practical sessions will take place in the classroom, and students are required to bring their laptops every lecture, and they are responsible on installing the tools that will be used through out the course.

Class participation and independent work. Students are expected to actively participate in all classes and allows perform independent work.

Class Etiquette. Please keep all cell phones and other electronic devices turned off during class. If your activities during class are deemed disruptive, you will be asked to leave. Use of a personal computer during class is prohibited except for note taking with Instructor permission.

Students Responsibility

Ritaj and Facebook: official communication through Ritaj. Students are assumed to check Ritaj several times a day. A Facebook Group is created for (*informal*) communication:

<https://www.facebook.com/groups/577156732420027/>

Attendance. Attendance is mandatory. University regulations regarding this matter will be strictly enforced.

Academic Honesty. Individual work must be each student's own work. Plagiarism or cheating will result in official University disciplinary review.

Missed Exams. There are no makeup exams.